NRES 746, Week 11, Lecture 2: Two Cores from One Tree

Answer key. Answers are in magenta sans-serif type; longer answers have a bar in the margin.

Instructions: Work alone (about 5 minutes), then compare with a neighbor. Keep laptops closed.

You want the mean annual ring width of the pinyon pines in a stand. Each increment core gives one measurement X with variance $\sigma^2 = 0.04$ mm^2 ($\sigma = 0.2$ mm). Some rules for variances and covariances that may come in handy (a and b are constants):

$$\mathrm{Var}(aX) = a^2\,\mathrm{Var}(X) \qquad \mathrm{Cov}(aX, bY) = ab\,\mathrm{Cov}(X, Y)$$

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X, Y)$$

where $\mathrm{Cov}(X, Y) = \rho\,\sigma_X\sigma_Y$ and ρ is the correlation between X and Y.

You average two cores, $M = \frac{1}{2}(X_1 + X_2)$. Taking the constant out first,

$$\mathrm{Var}(M) = \tfrac{1}{4}\,\mathrm{Var}(X_1 + X_2) = \tfrac{1}{4}\big[\mathrm{Var}(X_1) + \mathrm{Var}(X_2) + 2\,\mathrm{Cov}(X_1, X_2)\big]$$

1a. If the two cores come from different trees ($\rho = 0$), $\mathrm{Var}(M) =$ 0.02 mm^2.

1b. If both cores come from the same tree ($\rho = 0.8$), $\mathrm{Var}(M) =$ 0.036 mm^2.

$$\mathsf{Var}(M) = \tfrac{1}{4}\left(\sigma^2 + \sigma^2 + 2\rho\sigma^2\right) = \tfrac{1}{4}(0.04 + 0.04 + 2 \times 0.8 \times 0.04) = 0.036$$

Worked example from class: ten cores

Design A: ten cores from one tree (every pair has $\rho = 0.8$). Design B: one core from each of ten trees (independent). Which design estimates the stand's mean more precisely, and by how much? Ten cores from one tree are worth about how many independent cores?

More precise: Design B, by a factor of about 8.2. Ten cores from one tree $\approx$ 1.2 independent cores.

The variance of a sum is the sum of all 10 variances and all $10 \times 9 = 90$ pairwise covariances:

$$\mathsf{Var}(\bar{X}_A) = \frac{10(0.04) + 90(0.8)(0.04)}{100} = 0.0328 \qquad \mathsf{Var}(\bar{X}_B) = \frac{10(0.04)}{100} = 0.004$$

Design B is about 8.2 times more precise (variance 0.004 vs. 0.0328). Ten cores from one tree are worth about $0.04/0.0328 \approx 1.2$ independent cores. (They also measure that one tree, not the stand.)